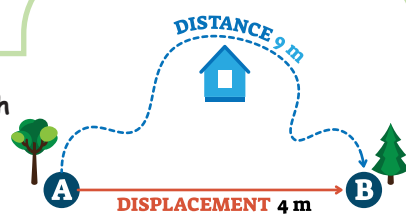


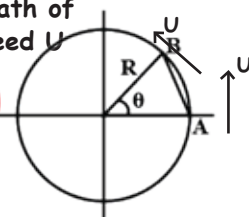


- Distance = Length of actual path
- Displacement = Length of shortest path
- Distance $\geq$ |displacement|



A particle moves from A to B in a circular path of radius R covering an angle θ with uniform speed U

- Distance = $\widehat{AB} = R\theta$
- Displacement = $AB = 2R\sin\left(\frac{\theta}{2}\right)$
- Ratio of Displacement to Distance = $\frac{\sin\left(\frac{\theta}{2}\right)}{\frac{\theta}{2}}$
- Time $t = \frac{R\theta}{U}$
- Average Velocity = $\frac{2U\sin\left(\frac{\theta}{2}\right)}{\frac{\theta}{2}}$
- Average Acceleration = $\frac{U^2\sin\left(\frac{\theta}{2}\right)}{R\frac{\theta}{2}}$



For uniform motion

Displacement = velocity $\times$ time
Average speed = |average velocity| = |instantaneous velocity|

Time average speed

$$v_{av} = \frac{\text{Total distance covered}}{\text{Total time elapsed}} = \frac{s_1 + s_2 + s_3 + \dots + s_n}{t_1 + t_2 + t_3 + \dots + t_n} = \frac{v_1 t_1 + v_2 t_2 + v_3 t_3 + \dots}{t_1 + t_2 + t_3 + \dots}$$

If $t_1 = t_2 = t_3 = \dots = t_n$

then

$$v_{av} = \frac{v_1 + v_2 + v_3 + \dots + v_n}{n}$$

for v_1 & v_2 ,

$$v_{av} = \frac{v_1 + v_2}{2} \text{ (Arithmetic mean of speeds)}$$

Distance average speed

$$v_{av} = \frac{\text{Total distance covered}}{\text{Total time elapsed}} = \frac{s_1 + s_2 + s_3 + \dots + s_n}{t_1 + t_2 + t_3 + \dots + t_n} = \frac{\frac{s_1}{v_1} + \frac{s_2}{v_2} + \frac{s_3}{v_3} + \dots + \frac{s_n}{v_n}}{\frac{s_1}{v_1} + \frac{s_2}{v_2} + \frac{s_3}{v_3} + \dots + \frac{s_n}{v_n}}$$

If $s_1 = s_2 = s_3 = \dots = s_n$

then

$$v_{av} = \frac{1}{\frac{1}{v_1} + \frac{1}{v_2} + \frac{1}{v_3} + \dots + \frac{1}{v_n}} \text{ for } v_1 \text{ \& } v_2, \quad v_{av} = \frac{2v_1 v_2}{v_1 + v_2} \text{ (Harmonic mean of speeds)}$$

Instantaneous Velocity $v = \frac{dx}{dt}$ $\Delta x = \int v dt$

Instantaneous Acceleration $a = \frac{dv}{dt}$ $\Delta v = \int a dt$

Case 1

$v = f(t)$ or $x = f(t)$

$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

Case 2

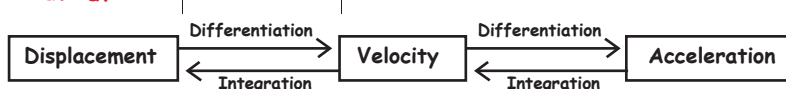
$V = f(x)$

$$a = V \frac{dV}{dx}$$

Case 3

$t = f(x)$

$$a = -(\text{double diff. of } t \text{ w.r.t. } x) \times V^3$$



Motion with constant acceleration: Equations of motion

$$(i) \quad v = u + at$$

$$(ii) \quad s = ut + \frac{1}{2} at^2$$

- A Person travels from A to B covers unequal distances in equal interval of time with constant acceleration a then

$$\text{initial velocity } U = \frac{3s_1 - s_2}{2t}$$

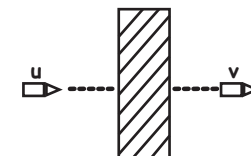
$$\text{Acceleration } a = \frac{s_2 - s_1}{t^2}$$



$$(iii) \quad v^2 = u^2 + 2as$$

- The number of planks required to stop the bullet

$$N = \frac{u^2}{u^2 - v^2}$$



- The two ends of a train moving with constant acceleration pass a certain point with velocities u and v. The velocity with which the middle point of the train passes the same point is

$$v_{Mid} = \sqrt{\frac{u^2 + v^2}{2}}$$



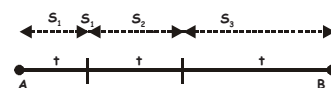
- Calculation of stopping distance $s = \frac{u^2}{2a}$



$$(iv) \quad s_n = u + \frac{a}{2}(2n-1)$$

- Ratio of distance travelled in equal interval of time in a uniformly accelerated motion from rest

$$s_1 : s_2 : s_3 = 1 : 3 : 5$$



- for uniform accelerated motion $v_{av} = \frac{u+v}{2}$

$$v_{av} = \frac{u+v}{2}$$

Different Cases	v-t graph	s-t graph
1. Uniform motion		
2. Uniformly accelerated motion with u = 0 at t = 0		
3. Uniformly accelerated with u ≠ 0 at t = 0 & s = 0 at t = 0		
4. Uniformly accelerated motion with u ≠ 0 and s = s0 at t = 0		
5. Uniformly retarded motion till velocity becomes zero		
6. Uniformly retarded then accelerated in opposite direction		

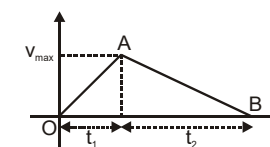
Important points about graphical analysis of motion

- Instantaneous velocity is the slope of position-time curve $\left[v = \frac{dx}{dt} \right]$
- Area of v-t curve gives displacement $\left[\Delta x = \int v dt \right]$
- Slope of velocity-time curve = instantaneous acceleration $\left[a = \frac{dv}{dt} \right]$
- Area of a-t curve gives change in velocity $\left[\Delta v = \int a dt \right]$

A car accelerates from rest at a constant rate α for some time, after which it decelerates at a constant rate β , to come to rest. If the total time elapsed is t, then

$$v_{max} = \frac{\alpha\beta}{\alpha+\beta} t$$

$$\text{Total Distance} = \frac{1}{2} \left(\frac{\alpha\beta}{\alpha+\beta} \right) t^2$$



MOTION UNDER GRAVITY

Sign Convention

(i) initial velocity

- +ve = upward motion
- ve = downward motion

(ii) Acceleration

Always -ve

(iii) Displacement

- +ve = final position is above initial position
- ve = final position is below initial position
- Zero = final position & initial position are at same level

- Object is dropped from top of a tower

(i) Ratio of displacement in equal interval of time $s_1 : s_2 : s_3 : \dots = 1 : 3 : 5 : \dots$

(ii) Ratio of time of covering equal distance

$$t_1 : (t_2 - t_1) : (t_3 - t_2) : \dots : (t_n - t_{n-1}) = 1 : (\sqrt{2} - \sqrt{1}) : (\sqrt{3} - \sqrt{2}) : \dots : (\sqrt{n} - \sqrt{n-1})$$

(iii) Ratio of total distance covered at the end of time $t : 2t : 3t : \dots = 1^2 : 2^2 : 3^2 : \dots$

- If a body is thrown vertically up with a velocity u in the uniform gravitational field (neglecting air resistance) then

(i) Maximum height attained $H = \frac{u^2}{2g}$

(ii) Time of ascent = time of descent $\frac{u}{g}$

(iii) Total time of flight = $\frac{2u}{g}$

(iv) Velocity of fall at the point of projection = u (downwards)

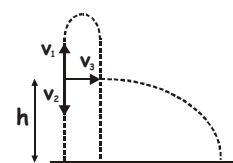
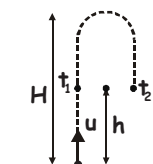
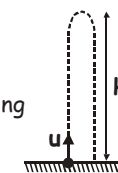
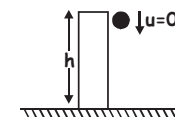
- At any point on its path the body will have same speed for upward journey and downward journey. If a body thrown upwards crosses a point in time t_1 & t_2 respectively then

$$\text{height of point } h = \frac{1}{2} g t_1 t_2 \quad \text{Maximum height } H = \frac{1}{8} g (t_1 + t_2)^2$$

$$\text{Time of flight} = t_1 + t_2 = \frac{2u}{g}$$

- A body is thrown upward, downward & horizontally with same speed takes time t_1 , t_2 & t_3 respectively to reach the ground then

$$t_3 = \sqrt{t_1 t_2} \quad \text{\& height from where the particle was throw is } h = \frac{1}{2} g t_1 t_2$$



MOTION ALONG A STRAIGHT LINE